EMPUF: A Leakage-Controlled Pre-Fabrication Benchmark for PCB Spectral Fingerprints

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Abstract

Printed circuit board (PCB) supply-chain authentication needs inexpensive physical fingerprints that can be inspected before a design is fabricated at scale. Electromagnetic (EM)-domain physical unclonable functions (PUFs) and radio-frequency (RF) fingerprints can derive entropy from microstrip geometry and fabrication variation, but their reported modeling resistance is easy to confound with encoding and train/test protocol choices. This paper presents EMPUF, an automated Ansys High Frequency Structure Simulator (HFSS) benchmark for side-coupled microstrip resonator PCB spectral fingerprints. EMPUF synthesizes tolerance-aware simulated populations with HFSS Optimetrics, extracts S_{11} spectral features, locks median thresholds on enrollment devices, and evaluates bit-index, design-only, metrology-assisted, learned-spectral, and white-box attackers under device-level splits. Across 100-instance HFSS populations with $N = 4, 8, 12$, and 16, exploratory results show that attack conclusions depend strongly on access to device-specific information. In a predeclared $N = 8$ split, bit-index/design-only, realized-geometry, and learned spectral attacks achieve 49.91%, 56.61%, and 75.71% accuracy, respectively. These results characterize simulated populations rather than deployment reliability.

CCS Concepts

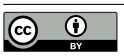
• **Security and privacy** → **Hardware security implementation**;
• **Hardware** → *Electromagnetic interference and compatibility*.

Keywords

physical unclonable functions, RF fingerprints, electromagnetic simulation, hardware security, pre-fabrication benchmarking

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1 Introduction

Printed circuit board (PCB) supply-chain anti-counterfeiting needs authentication primitives that are unique per device, hard to clone, and cheap enough for board-level deployment. Physical unclonable functions (PUFs) target this goal by deriving fingerprints from manufacturing variation [3], but most mature designs are silicon-based (static random-access memory (SRAM) [4], ring-oscillator [14], arbiter [6]) and therefore mismatched to printed or board-level security markers.

Electromagnetic (EM)-domain PUFs offer a printed alternative: entropy can arise from resonator geometry, PCB substrate properties, and etching/lamination variation. The radio-frequency (RF) input-reflection coefficient S_{11} of a passive microstrip resonator array provides a rich fingerprint for each physical instance [16], while FR-4 fabrication keeps cost within radio-frequency identification (RFID) budgets. For such structures, simulation enables controlled pre-fabrication screening but does not establish the reliability of a deployed physical device.

The missing piece is automation and evaluation discipline: existing EM-PUF demonstrations are largely manual, single-design studies [2, 11], whereas silicon PUFs benefit from systematic design and attack-evaluation flows [15]. EM-PUF designers still lack a flow that turns RF and benchmark constraints into layouts, simulates tolerance-driven populations, locks encoding thresholds without test-device leakage, and evaluates multiple attacker feature sets under a reproducible protocol.

This paper presents EMPUF, an Ansys High Frequency Structure Simulator (HFSS)-based pre-fabrication benchmark for side-coupled microstrip resonator EM-PUFs. The contributions are:

- An automated pipeline that converts parameterized HFSS layouts, Optimetrics-driven Monte Carlo samples, and S_{11} sweeps into reproducible PCB spectral-fingerprint datasets.
- A leakage-controlled protocol with enrollment-only median thresholds, device-level splits, and explicit attacker information levels.
- A controlled comparison of how realized-geometry and spectral access, resonator count, feature choice, and split granularity affect attack accuracy.

2 Background and Related Work

2.1 EM-Domain Physical Unclonable Functions

An EM-PUF is a printed array of N microstrip resonators excited through a feed line. Each resonator behaves as a weakly coupled resistance–inductance–capacitance (RLC) tank whose resonant frequency f_{res}, notch depth d, quality factor Q, and phase slope ϕ'

Table 1: Positioning relative to adjacent RF/PUF research.

Work type	Pre-fab design	Full-wave MC	Attack split
Silicon PUF	Yes	No	device/corner
Manual EM-PUF	Limited	Limited	often absent
RFFI	No	No	emitter/device
EMPUF	Yes	Yes	device-level

depend on geometry and substrate properties. Fabrication tolerances perturb these spectral features, and a 3–8 GHz S_{11} sweep captures all N signatures. We encode the resulting $4N$ features using median thresholds locked on enrollment devices, which makes the response depend on ordering rather than absolute calibration while avoiding test-device participation in threshold selection.

2.2 Related Work

Design automation. Silicon PUF flows evaluate SRAM, delay, or configurable ring-oscillator designs under process–voltage–temperature (PVT) and attack models [13, 15], but they do not address antenna layout or full-wave EM simulation. HFSS automation has recently accelerated RF design [1]; EMPUF uses the same toolchain to make modeling-resistance evaluation a first-class benchmark objective, alongside RF performance.

EM signatures for authentication. NotchPUF is the closest PCB-level predecessor: it uses microstrip notch filters on fabricated boards and reports repeated-measurement PUF metrics [7]. Other EM-PUFs demonstrate resonant-cavity or photonic-assisted fingerprints [2, 16], while radio-frequency fingerprint identification (RFFI) identifies existing radios post-fabrication [10, 17, 18]. EMPUF differs from NotchPUF by automating pre-fabrication HFSS population generation and by calibrating encoding/splitting/attacker protocols; it does not replace the fabricated-board validation that NotchPUF already demonstrates. Coupled ring-oscillator PUFs (RO-PUFs) [8] also motivate future exploration of spatial coupling among EM resonators. This distinction is important experimentally: RFFI optimizes classifier identification accuracy over existing emitters, whereas EM-PUF design must optimize uniqueness, uniformity, reliability, and resistance to attackers who know the design but not each fabricated instance's tolerance realization. Table 1 summarizes these differences in design stage, full-wave Monte Carlo (MC) scope, and attack-evaluation split.

2.3 Threat Model

We treat the present structure as a weak-PUF or static PCB spectral fingerprint: one fabricated device yields a fixed response vector from a fixed S_{11} sweep. It is therefore not a strong PUF with a large same-device challenge space. We assume Kerckhoffs' principle: the attacker knows the layout generator, encoding rule, classifier family, and tolerance ranges. What varies is the attacker's access to the target device's realized physical or spectral state. We distinguish three device-information levels: a nominal-layout baseline, metrology-assisted access to realized geometry, and access to measured spectral features through a machine-learning (ML) attacker. The bit identifier (Bit-ID) is retained as a response-bias baseline. A deterministic white-box replay of the known encoder is used only as an implementation check. Section 4 defines the corresponding feature matrices.

3 EMPUF Design and Encoding

Figure 1 summarizes the four EMPUF stages: design generation, Monte Carlo population synthesis, feature extraction/encoding, and benchmark evaluation.

3.1 Feature Extraction and Encoding

Using scikit-rf [12], each detected resonator contributes f_{res}, d, Q, and ϕ', yielding $\mathbf{F} \in \mathbb{R}^{M\times N\times 4}$.

For encoding, we adopt **locked enrollment-median thresholding**: for each (feature, resonator) pair, a threshold is computed only from the enrollment devices in the current split. Development and test devices are then encoded with the fixed threshold. Formally, for feature k and resonator j:

$$b_{j,k}^{(i)} = \mathbf{1}\left[F_{i,j,k} > \tau_{j,k}\right], \quad \tau_{j,k} = \text{median}_{m\in\mathcal{D}_{\text{enroll}}} F_{m,j,k}. \tag{1}$$

This avoids transductive threshold selection from the held-out devices. Bits with non-finite feature values are encoded as 0 in the benchmark scripts and reported in each run's encoding metadata.

3.2 Resonator Geometry and Population Synthesis

Our baseline cell is a U-shaped microstrip resonator on a 1.6 mm FR-4 substrate ($\varepsilon_r = 4.4$, $\tan\delta = 0.02$). Each resonator $r \in \{1,\dots,N\}$ has two Monte Carlo variables (Fig. 2): side-arm length l_r (primary control of f_{res}) and feed-to-resonator gap g_r (notch-depth control). The feed line is a continuous 50 Ω microstrip trace ($w_{\text{feed}} = 3.0$ mm). The resonator span, arm width ($w_{\text{arm}} = 0.45$ mm), and nominal pitch (8 mm in the linear layouts) are fixed layout constants; only l_r and g_r are perturbed during population synthesis.

3.2.1 Initial Placement Heuristic. To achieve uniform coverage of the 3–8 GHz band, we place N resonators with target resonant frequencies spaced at equal intervals:

$$f_r^{\text{target}} = f_{\min} + \frac{r-1}{N-1}(f_{\max} - f_{\min}), \quad r = 1,\dots,N, \quad N > 1 \tag{2}$$

where $f_{\min} = 3.2$ GHz and $f_{\max} = 7.4$ GHz. Quarter-wave scaling provides the physical initialization trend, $l_r \approx c/(4f_r^{\text{target}}\sqrt{\varepsilon_{r\text{eff}}})$, with effective permittivity $\varepsilon_{r\text{eff}}$ from microstrip transmission-line theory. The implemented generator then interpolates calibrated $N = 8$ reference lengths at the target frequencies. A guided notch-placement optimization then adjusts each resonator's nominal parameters with objectives of $d > 5$ dB and inter-resonator frequency spacing $> 3\times$ the 3 dB bandwidth. Monte Carlo perturbations may move individual instances outside these nominal targets.

3.2.2 Monte Carlo Population Synthesis. The Monte Carlo engine reuses one baseline HFSS project through the Python interface to Ansys Electronics Desktop (PyAEDT), as summarized in Algorithm 1. For each instance, it samples $\Delta l_r \sim \mathcal{N}(0, 50\,\mu\text{m})$ and $\Delta g_r \sim \mathcal{N}(0, 15\,\mu\text{m})$, truncates both to $\pm 3\sigma$ manufacturable ranges, updates the model, runs a 3–8 GHz 501-point interpolating sweep, and exports S_{11}.

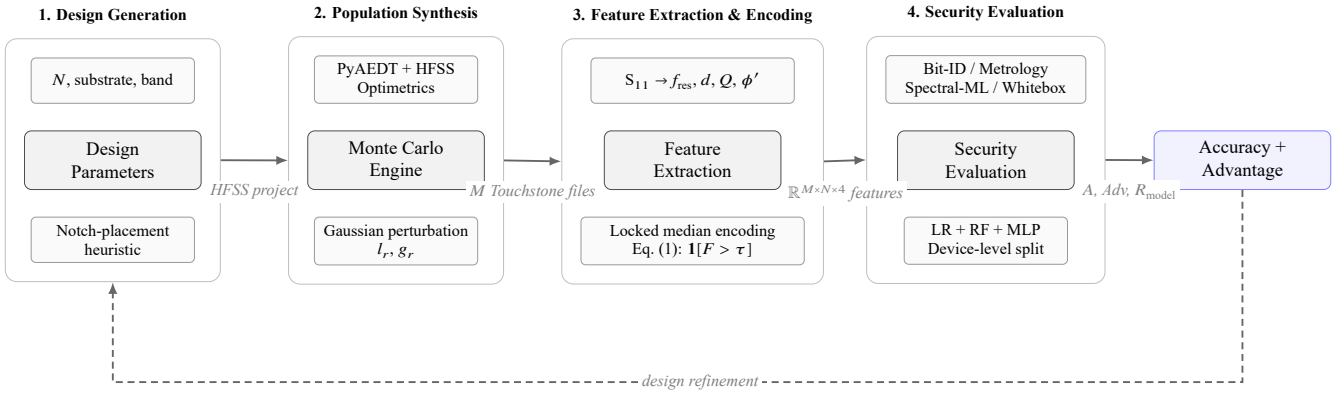


Figure 1: EMPUF framework overview: automated HFSS design generation → Monte Carlo population synthesis → feature extraction and median encoding → multi-threat-model evaluation. A dashed feedback loop enables iterative design refinement guided by benchmark scores before fabrication.

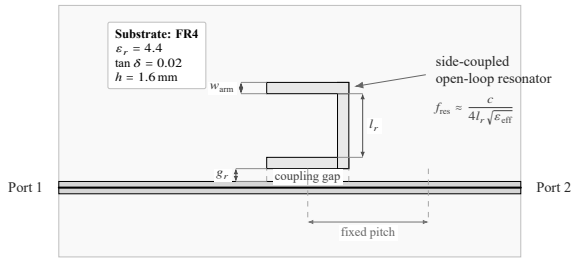


Figure 2: Side-coupled open-loop microstrip resonator geometry used in HFSS. l_r is the side-arm length and g_r is the feed-to-resonator coupling gap. The feed width, arm width, and resonator pitch are fixed layout constants.

Algorithm 1 Monte Carlo Population Synthesis

Require: Baseline HFSS project P_0, population size M
Ensure: Touchstone files $\{S_{11}^{(1)}, \ldots, S_{11}^{(M)}\}$
  Sample M truncated-Gaussian points in $\mathbb{R}^{2N}$
  for $i = 1$ **to** M **do**
    Update all $\{l_r, g_r\}_{r=1}^{N}$ in HFSS
    Run 3–8 GHz interpolating sweep and export $S_{11}^{(i)}$
  end for

3.3 Scaling to Multiple Configurations

For the N-sweep, N is a top-level variable and the generator recomputes target frequencies, geometries, and perturbation ranges automatically.

3.4 Automation Scope

The flow takes resonator count, substrate and band settings, tolerance distributions, population size, and random seed as configuration inputs. It then generates the HFSS project, samples and logs each geometry realization, runs the population sweep, exports Touchstone files, extracts spectral features, fits enrollment thresholds, evaluates attacks, and regenerates the paper tables and figures. Only project generation and full-wave population simulation require HFSS/PyAEDT. All downstream stages run from the exported Touchstone, comma-separated value (CSV), and JavaScript Object Notation (JSON) files. Each run records sampled variables, split seeds, thresholds, and classifier settings. Table 2 summarizes the solver boundary and replay requirements.

Table 2: Automated flow boundary and replay requirements.

Stage	Automated output	Needs
Design/population	HFSS project, geometry log, S_{11} sweeps	HFSS
Encoding	Feature table, thresholds, response bits	Python
Evaluation	Fixed splits, attacks, confidence intervals, figures	Python

4 Experimental Protocol

4.1 Classifier Models

We use three scikit-learn [9] classifiers with complementary biases:

- **Logistic regression (LR)**: ℓ_2 regularization with class-balanced weights and StandardScaler preprocessing, testing linearly decodable leakage.
- **Random forest (RandF)**: 200 class-balanced trees capturing axis-aligned nonlinear interactions among geometry and bit identity.
- **Multilayer perceptron (MLP)**: hidden sizes 64/32, rectified linear unit (ReLU) activations, early stopping, and $\alpha = 10^{-4}$, testing smooth nonlinear leakage.

Pipelines use median imputation for rare missing resonator features.

4.2 Device-Level Train/Test Splitting

A critical requirement is **device-level splitting**. Because each device contributes B bits, bit-pair splitting can place the same device in both train and test sets, leaking its EM identity. We instead

partition device indices:

$$\mathcal{D}_{\text{train}} \cap \mathcal{D}_{\text{test}} = \emptyset, \quad \text{all bits of device } d \in \mathcal{D}_{\text{train}} \rightarrow \text{train set} \quad (3)$$

4.3 Threat Model Feature Construction

For each threat model, we construct a feature matrix $\mathbf{X} \in \mathbb{R}^{MB \times D}$ and target vector $\mathbf{y} \in \{0,1\}^{MB}$:

- **Bit-ID**: $\mathbf{X}$ contains only the bit ID one-hot encoding, with $D = B$. This measures response bias under the locked thresholds.
- **Design**: the attacker knows the nominal layout. Since the nominal layout is constant across devices in one configuration, Design has the same feature matrix and numerical result as Bit-ID in these experiments.
- **Metrology**: $\mathbf{X}$ concatenates the realized sampled geometry ($2N$ values: stub length and gap width per resonator from the Monte Carlo variable log) with the bit ID one-hot encoding, $D = 2N + B$. This is stronger than nominal-design access and models metrology-assisted access.
- **Spectral-ML**: $\mathbf{X}$ concatenates the full extracted feature tensor $\mathbf{F}$ (flattened to $4N$ values) with the bit ID one-hot encoding, $D = 4N + B$. LR, RandF, and MLP learn the spectral-to-response mapping without direct access to the threshold rule.
- **White-box**: the attacker directly applies Eq. 1 with the measured features and locked thresholds, deterministically recovering the encoded response. It serves as an encoder implementation check.

4.4 Evaluation Protocol

The N-sweep uses 5 independent device-level splits for each train fraction in $\{0.3, 0.5, 0.7\}$. In each split, thresholds are computed from the train/enrollment devices only, then applied unchanged to the test devices. The maximum over classifiers and splits is an exploratory sensitivity statistic.

The confirmatory $N = 8$ experiment uses a locked calibration/development/test protocol (Fig. 3). The calibration set (30 devices) fits median thresholds only. The development set (35 devices) selects LR, RandF, or MLP and trains the selected attacker. The final-test set (35 devices) is evaluated once. Confidence intervals (CIs) use a device-level bootstrap: final-test devices are resampled with replacement and all bits from each sampled device are kept together. We report accuracy and

$$\text{Adv} = 2\left(\max(A, 1-A) - 0.5\right), \quad (4)$$

where A is accuracy. This symmetric statistic assigns equal advantage to perfect prediction and perfect inversion. Table 3 summarizes the complete evaluation protocol.

5 Simulation Results

5.1 Simulation Setup

We simulate 100-instance populations for $N \in \{4, 8, 12, 16\}$ in HFSS 2024 R2 using 3–8 GHz, 501-point sweeps on 1.6 mm FR-4. Length/gap tolerances are $\sigma_l = 50\ \mu$m and $\sigma_g = 15\ \mu$m. Fair scaling retains all $4N$ feature-resonator candidates, and median thresholds

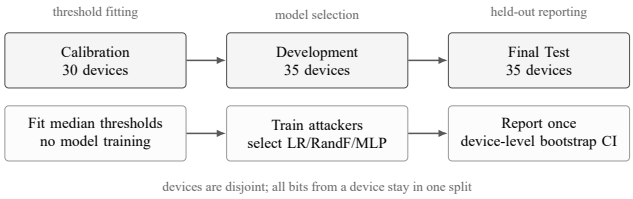


Figure 3: Locked final protocol used for the N=8 confirmation experiment. Thresholds, model selection, and final reporting use disjoint device sets.

Table 3: Benchmark-evaluation protocol summary.

Item	Setting
Split	Device-level; no bits from a device cross split boundaries
Thresholds	Median thresholds fit on train/enrollment or calibration devices only
Threats	Bit-ID, Design, Metrology, Spectral-ML, and White-box
Models	LR, RandF, and MLP scikit-learn pipelines
Sensitivity sweep	Train fractions 0.3, 0.5, 0.7; 5 seeds
Final check	30 calibration / 35 development / 35 final-test devices
Metrics	Accuracy, device-bootstrap 95% CI, and Adv

Table 4: Fair scaling encoding summary. One bit is retained for each feature-resonator pair before missing-feature handling.

N	$\lvert f_{\text{res}}\rvert$	$\lvert d\rvert$	$\lvert Q\rvert$	$\lvert \phi'\rvert$	Total
4	4	4	4	4	16
8	8	8	8	8	32
12	12	12	12	12	48
16	16	16	16	16	64

are locked on the train/enrollment devices in each split. The ablation study uses the same fair $N = 8$ encoding. Table 4 reports the retained feature counts.

5.2 Modeling Accuracy Scaling with Resonator Count

Table 5 and Fig. 4 report attack accuracy across resonator counts.

Because nominal geometry is constant within each configuration, Design and Bit-ID use the same inputs; their identical results are a baseline property. Metrology uses the sampled l_r/g_r values and peaks at 61.25% for $N = 8$, showing modest device-specific signal in realized geometry. Spectral-ML reaches 77.71–90.83%, whereas deterministic White-box replay recovers the encoded bits by construction. The exploratory maxima compare attack surfaces; confirmatory claims use the locked final test below.

Table 5: Locked-threshold attack results by resonator count. Max accuracy is computed across LR, RandF, and MLP; train fractions (0.3, 0.5, 0.7); and five seeds for learned attacks. White-box is the deterministic threshold rule.

N	Bit-ID/Design	Metrology	Spectral-ML	White-box
4	59.25%	58.96%	90.83%	100.00%
8	57.95%	61.25%	86.67%	100.00%
12	56.67%	56.88%	78.33%	100.00%
16	55.69%	55.83%	77.71%	100.00%

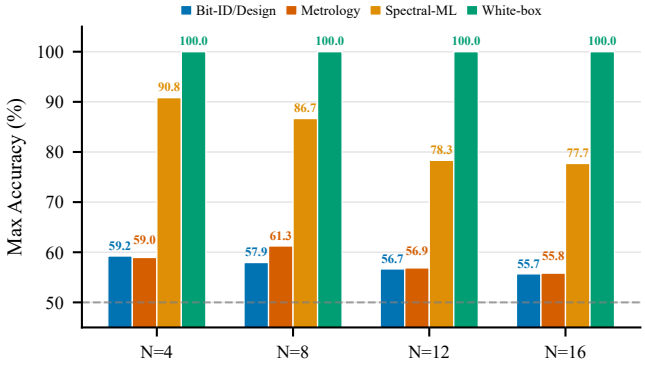


Figure 4: Attack accuracy vs. resonator count N under locked train-median encoding. Design-only access collapses to the bit-index baseline; realized geometry is a metrology-assisted threat; learned spectral attacks are strong; and white-box threshold access exactly reconstructs responses.

Table 6: N=8 final locked protocol with device-bootstrap 95% confidence intervals. Models are selected on development devices only, then evaluated once on final-test devices.

Threat	Model	Final acc. (95% CI)	Adv (95% CI)
Bit-ID / Design	MLP	49.91% [47.77, 52.41]	0.002 [0.000, 0.052]
Metrology	RandF	56.61% [52.86, 60.63]	0.132 [0.057, 0.213]
Spectral-ML	RandF	75.71% [72.32, 79.02]	0.514 [0.446, 0.580]
White-box	Rule	100.00% [100.00, 100.00]	1.000 [1.000, 1.000]

5.3 Locked Final Test

Table 6 reports the predeclared $N = 8$ final check from Fig. 3. The bit-index and design-only baselines collapse to chance on the held-out final devices, while Metrology remains weakly predictive. The learned spectral attacker reaches 75.71% final accuracy (Adv=0.514), and White-box remains exactly 100%.

5.4 Stronger Spectral Attack Baselines

Table 7 compares stronger S_{11}-access baselines on the same split. The RandF trace attacker uses 128-point interpolated magnitude and 128-point unwrapped phase plus bit ID. The one-dimensional convolutional neural network (1D-CNN) attacker uses 256-point magnitude/phase traces and a bit embedding; hyperparameters are selected on development devices only. The vector-fitting attacker fits each Touchstone trace with scikit-rf vector fitting, then uses

Table 7: N=8 S_{11}-access attacker baselines under the same final locked protocol. All intervals are device-bootstrap 95% CIs.

Input representation	Model	Final acc. (95% CI)	Adv (95% CI)
Extracted spectral	RandF	75.71% [72.32, 79.02]	0.514 [0.446, 0.580]
Raw S_{11}	RandF	70.54% [67.05, 74.11]	0.411 [0.341, 0.482]
Vector-fitted S_{11}	RandF	68.30% [65.54, 71.25]	0.366 [0.311, 0.425]
Raw S_{11}	1D-CNN	63.13% [60.36, 66.07]	0.263 [0.207, 0.321]

Table 8: Locked-threshold feature ablation results (N=8, Metrology threat model). Mean ± standard deviation and maximum are across 45 learned-attack runs.

Feature group	Bits	Mean acc.	Max acc.	Max Adv
Phase-only	8	52.07 ± 2.27%	56.67%	0.133
Magnitude-only	24	57.95 ± 2.06%	62.25%	0.245
Full	32	56.18 ± 2.04%	61.25%	0.225

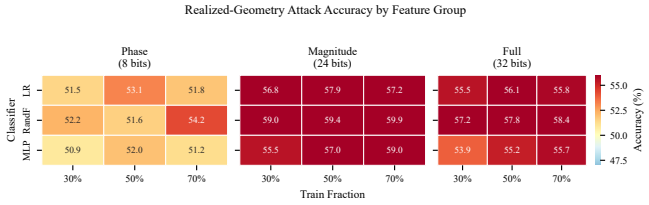


Figure 5: Locked-threshold Metrology attack accuracy heatmap: classifier × train fraction × feature group. Phase-only features consistently cluster near 50% (blue), while magnitude features show weak but consistent above-chance signal (light red).

sorted pole/residue coefficients, constant terms, and fitting error as features. RandF over downsampled raw traces is the strongest of these added baselines, reaching 70.54% [67.05, 74.11]. The CNN result is weaker on this 100-device population, which is consistent with a small-data regime rather than evidence that deep trace models are harmless. Vector fitting is intermediate at 68.30% [65.54, 71.25] with median root-mean-square (RMS) fit error 0.241. None of these baselines is a functional-cloning attack.

5.5 Protocol and Feature Sensitivity

5.5.1 Feature-Group Predictability. We ablate $N = 8$ into phase-only ϕ' (8 bits), magnitude-only (f_{res}, d, Q) (24 bits), and full (32 bits) encodings. Table 8 and Fig. 5 report the corresponding metrology-assisted attack results.

Phase-only accuracy averages 52.07% and peaks at 56.67%, lower than the magnitude-only and full encodings under the locked-threshold protocol. These bits appear less tied to the sampled length/gap variables and are therefore candidates for subsequent physical-reliability testing.

Magnitude-only bits are the most predictable group (max 62.25%, Adv=0.245), consistent with their direct dependence on resonant frequency, notch depth, and Q-factor.

The full encoding is intermediate. Under this attacker and sample size, phase bits are less predictable from sampled geometry than magnitude bits.

6 Discussion

6.1 Design Guidelines

The simulation benchmark gives two practical rules for pre-fabrication design exploration. First, resonator count, threshold policy, and split granularity must be reported together: locked train-median thresholds avoid the test-device participation hidden in global population medians. Second, nominal layout access is not device-specific when every device shares the same design; the meaningful geometry benchmark is metrology-assisted access to realized tolerance samples.

These rules also expose a threat-model caveat: an attack can look close to random when its inputs omit the post-fabrication variables that define the response. Therefore Bit-ID and Design are response-bias baselines, while Metrology and Spectral-ML are stronger diagnostics. Device-level train/test splits are necessary because each simulated device contributes multiple response bits; pair-level splitting would allow one device to appear in both training and test sets. This requirement is independent of EM-PUFs and applies to any benchmark that exposes multiple observations per physical device.

The benchmark also clarifies what must be reported for reproducibility: population size, retained-bit policy, threshold-fitting set, split granularity, attacker feature access, and whether spectral access is learned or white-box. Without these details, two EM-PUF designs can appear to differ in modeling resistance when the difference is actually caused by encoding or evaluation choices.

6.2 Limitations and Future Work

Our claims are bounded by four limitations. First, the study is HFSS-based; measurements of fabricated boards with a vector network analyzer (VNA) are needed before deployment. Machine-learning-assisted RF design work motivates automated design exploration [1], but it does not replace an EM-PUF validation campaign. Such measurements should quantify uniqueness, uniformity, bit error rate, repeatability over repeated reads, temperature drift, connector perturbation, reader distance/angle sensitivity, and VNA calibration/noise effects. Second, the Monte Carlo population contains 100 simulated instances per configuration; larger populations and fabrication-lot variation are needed to estimate tails and enrollment yield. Third, the sweep covers $N \in \{4, 8, 12, 16\}$ in a fixed 3–8 GHz band; wider bands, re-spaced targets, and intentionally coupled resonators remain open directions, motivated by coupled silicon PUFs [8]. Fourth, the evaluated LR, RandF, MLP, 1D-CNN, and vector-fitting baselines do not exhaust all adversaries. Larger trace models, sequence models, equivalent-circuit synthesis, and active RF replay [5] remain open. Functional cloning is not evaluated: a future adversary may synthesize a matching RF response without reproducing the microscopic manufacturing state.

7 Conclusion

EMPUF provides an automated pre-fabrication benchmarking flow for PCB spectral fingerprints by combining Monte Carlo population synthesis, locked-threshold median encoding, and device-level attack evaluation. The central lesson is that reported EM-PUF modeling resistance depends as much on encoding, threshold selection, and attacker feature access as on RF geometry. The framework and datasets establish a reproducible basis for screening evaluation choices and candidate encodings before fabrication.

Acknowledgments

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